

# Run report — c5\_confirm\_seed101

Generated : 2026-08-03 15:21:32  
Primary metric : 0.5035 (complete364\_macro\_f1)  
Majority ref : 0.4180 margin +0.0855  
Best previous : c1\_nested\_ensemble @ 0.5327  
Verdict : no improvement (-0.0292)  
Duration : 281s (4.7 min)

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## HEADLINE METRICS

|                                 |        |
|---------------------------------|--------|
| complete364_macro_f1            | 0.5035 |
| complete364_macro_f1_single_ref | 0.5155 |
| mean_chosen_k                   | 3.0000 |
| fixed_k                         | True   |
| splits_seed                     | 101    |
| complete364_weighted_f1         | 0.5953 |
| complete364_balanced_acc        | 0.5047 |
| complete364_accuracy            | 0.5892 |
| n_folds                         | 5      |
| subject_relative                | False  |

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## WHAT CHANGED

Iteration 5 - HELD-OUT CONFIRMATION of the c1 protocol (frozen: 7 original feature sets, 7 models, k=3 fixed)

# Per-fold outer results

| fold | n_test | chosen_k | ens_macro_f1 | ens_weighted_f1 | ens_balanced_acc | ens_accuracy | single_macro_f1 | single_weighted_f1 |
|------|--------|----------|--------------|-----------------|------------------|--------------|-----------------|--------------------|
| 0    | 84     | 3        | 0.5510       | 0.6380          | 0.5500           | 0.6429       | 0.5654          | 0.6142             |
| 1    | 77     | 3        | 0.4870       | 0.5477          | 0.4869           | 0.5455       | 0.5260          | 0.5454             |
| 2    | 63     | 3        | 0.4763       | 0.6179          | 0.4838           | 0.5873       | 0.5358          | 0.6397             |
| 3    | 77     | 3        | 0.4463       | 0.5221          | 0.4458           | 0.5195       | 0.4011          | 0.4685             |
| 4    | 63     | 3        | 0.5569       | 0.6508          | 0.5569           | 0.6508       | 0.5492          | 0.5953             |

## Inner-CV model ranking (mean over folds)

| features             | model      | inner_macro_f1 |
|----------------------|------------|----------------|
| video                | logreg_c01 | 0.5790         |
| video                | logreg     | 0.5657         |
| video                | hgb        | 0.5336         |
| audio+video          | logreg_c01 | 0.5308         |
| audio+video          | hgb        | 0.5245         |
| physfeat+video       | logreg     | 0.5230         |
| all                  | logreg_c01 | 0.5208         |
| physfeat+video       | logreg_c01 | 0.5197         |
| all                  | logreg     | 0.5174         |
| physfeat+audio+video | logreg     | 0.5161         |
| audio+video          | logreg     | 0.5144         |
| video                | extratrees | 0.5106         |
| physfeat+audio+video | logreg_c01 | 0.4990         |
| audio                | logreg_c01 | 0.4975         |
| physfeat+video       | hgb        | 0.4899         |

# Per-fold selection

| fold | chosen_k | inner_score_at_k | k_grid_scores |         |         |         |         |                                                    |  |  |  |
|------|----------|------------------|---------------|---------|---------|---------|---------|----------------------------------------------------|--|--|--|
| 0    | 3        | 0.5550           | 1:0.563       | 2:0.550 | 3:0.555 | 5:0.544 | 8:0.544 | video/logreg   video/logreg_c01                    |  |  |  |
| 1    | 3        | 0.6490           | 1:0.630       | 2:0.633 | 3:0.649 | 5:0.641 | 8:0.645 | video/logreg_c01   all/logreg_c01   audio+video/l  |  |  |  |
| 2    | 3        | 0.5540           | 1:0.569       | 2:0.556 | 3:0.554 | 5:0.540 | 8:0.509 | video/logreg_c01   video/logreg   al               |  |  |  |
| 3    | 3        | 0.6129           | 1:0.613       | 2:0.612 | 3:0.613 | 5:0.566 | 8:0.537 | video/logreg   video/logreg_c01                    |  |  |  |
| 4    | 3        | 0.5201           | 1:0.527       | 2:0.525 | 3:0.520 | 5:0.507 | 8:0.519 | video/logreg_c01   audio+video/logreg_c01   audio+ |  |  |  |

# Config

```
{
  "subject_relative": false,
  "ensemble_top_k": 3,
  "fixed_k": true,
  "feature_sets": [
    "videofeat",
    "videofeat+video",
    "videofeat+audio",
    "videofeat+physfeat+audio",
    "video",
    "audio",
    "physfeat",
    "physfeat+video",
    "audio+video",
    "physfeat+audio+video",
    "all",
    "all+videofeat"
  ],
  "models": [
    "logreg",
    "logreg_c01",
    "svc_rbf",
    "svc_lin",
    "rf",
    "extratrees",
    "hgb"
  ],
  "n_folds": 5,
  "data_tag": "full",
  "selection": "nested GroupKFold (inner selection on train subjects only)"
}
```